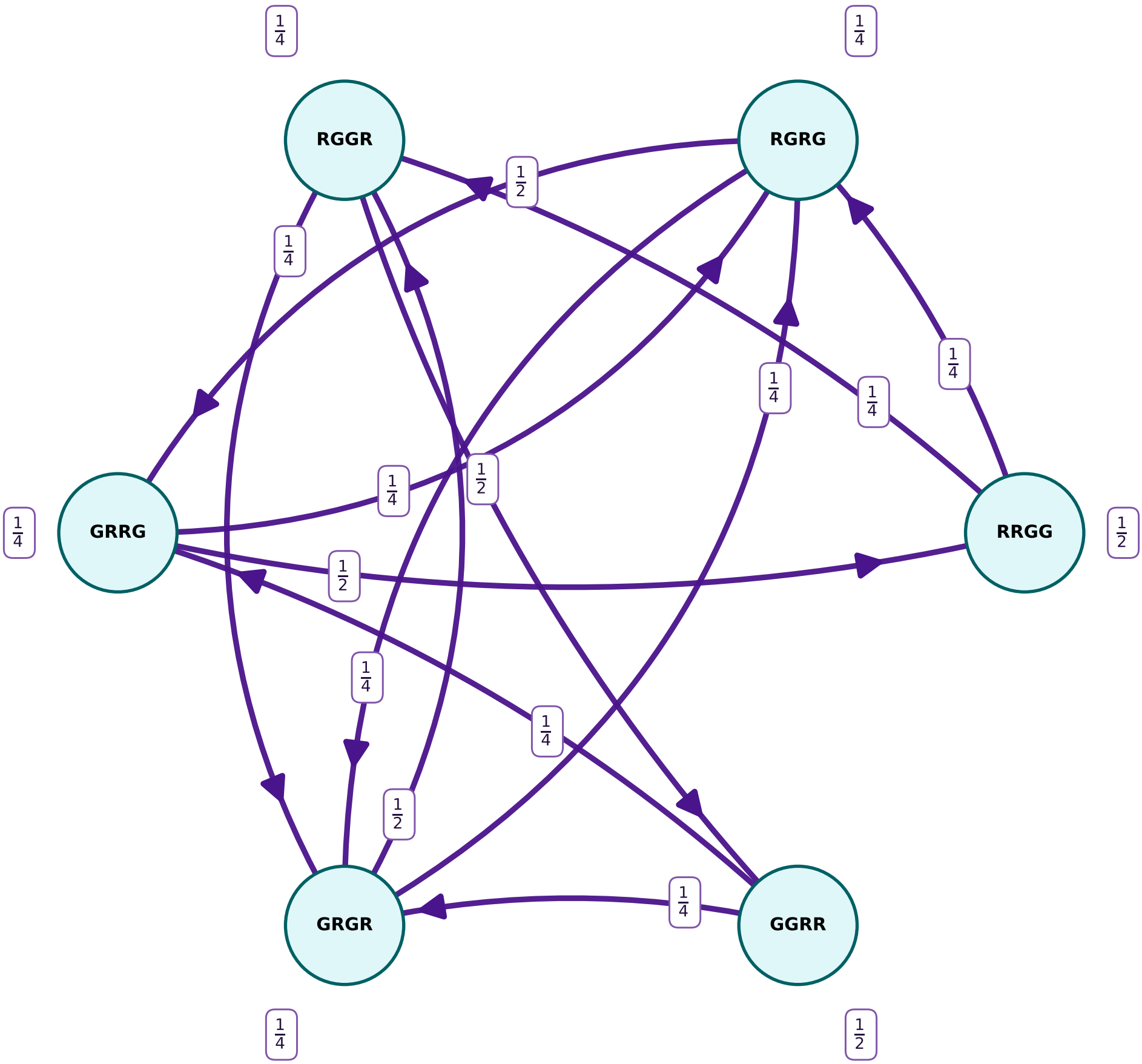Directed graph $\mathcal{G}_4$